Lab Manual – 4

Code/Course: 23CSE312/Distributed Systems

Topic: Message Passing Interface (MPI) on a localhost

[CO05] [BTL-3]
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Results

Question 1:

```python
from mpi4py import MPI

comm = MPI.COMM_WORLD
rank = comm.Get_rank()
size = comm.Get_size()

if size < 2:
    print("Run the program with at least 2 processes.")
    exit()

if rank == 0:
    print("Process 0: Sending 'ping' to Process 1")
    comm.send("ping", dest=1)
    message = comm.recv(source=1)
    print(f"Process 0: Received message from Process 1: {message}")

elif rank == 1:
    message = comm.recv(source=0)
    print(f"Process 1: Received message from Process 0: {message}")
    comm.send("pong", dest=0)
    print("Process 1: Sent 'pong' to Process 0")
```

```
❯ mpirun -n 2 python 1.py
Process 1: Received message from Process 0: ping
Process 1: Sent 'pong' to Process 0
Process 0: Sending 'ping' to Process 1
Process 0: Received message from Process 1: pong
```

Question 2:

```
from mpi4py import MPI

comm = MPI.COMM_WORLD
rank = comm.Get_rank()

if rank == 0:
    data = (100, [1, 2, 3], {"subject": "MPI"})
else:
    data = None

data = comm.bcast(data, root=0)
print(f"Process {rank} received: {data}")
```

```
❯ mpirun -n 4 python 2.py
Process 0 received: (100, [1, 2, 3], {'subject': 'MPI'})
Process 1 received: (100, [1, 2, 3], {'subject': 'MPI'})
Process 2 received: (100, [1, 2, 3], {'subject': 'MPI'})
Process 3 received: (100, [1, 2, 3], {'subject': 'MPI'})
```

Question 3:

```
from mpi4py import MPI

comm = MPI.COMM_WORLD
rank = comm.Get_rank()

if rank == 0:
    data = [(10, 'A'), (20, 'B'), (30, 'C'), (40, 'D')]
else:
    data = None

received = comm.scatter(data, root=0)
print(f"Process {rank} received: {received}")
```

```
❯ mpirun -n 4 python 3.py
Process 0 received: (10, 'A')
Process 1 received: (20, 'B')
Process 2 received: (30, 'C')
Process 3 received: (40, 'D')
```

Question 4:

```
from mpi4py import MPI

comm = MPI.COMM_WORLD
rank = comm.Get_rank()

local_data = (rank, rank ** 2)
all_data = comm.gather(local_data, root=0)
```

```
if rank == 0:
    print("Collected Data:", all_data)
```

```
❯ mpirun -n 4 python 4.py
Collected Data: [(0, 0), (1, 1), (2, 4), (3, 9)]
```

Question 5:

```
from mpi4py import MPI

comm = MPI.COMM_WORLD
rank = comm.Get_rank()

if rank == 0:
    comm.send("Data for Process 1", dest=1)
    comm.send("Data for Process 2", dest=2)
    comm.send("Data for Process 3", dest=3)
else:
    data = comm.recv(source=0)
    print(f"Process {rank} received: {data}")
```

```
❯ mpirun -n 4 python 5.py
Process 1 received: Data for Process 1
Process 2 received: Data for Process 2
Process 3 received: Data for Process 3
```

Question 6:

```
from mpi4py import MPI

comm = MPI.COMM_WORLD
rank = comm.Get_rank()

if rank == 0:
    role = "Manager"
elif rank == 1:
    role = "Worker A"
elif rank == 2:
    role = "Worker B"
elif rank == 3:
    role = "Worker C"

print(f"Process {rank} role: {role}")
```

```
❯ mpirun -n 4 python 6.py
Process 0 role: Manager
Process 3 role: Worker C
Process 2 role: Worker B
Process 1 role: Worker A
```

Question 7:

```
from mpi4py import MPI
```

```
comm = MPI.COMM_WORLD
rank = comm.Get_rank()

if rank == 0:
   data = ("MPI", {"type": "Parallel Programming", "year": 2025})
else:
   data = None

data = comm.bcast(data, root=0)
print(f"Process {rank} received: {data}")
```

```
❯ mpirun -n 4 python 7.py
Process 0 received: ('MPI', {'type': 'Parallel Programming', 'year': 2025})
Process 1 received: ('MPI', {'type': 'Parallel Programming', 'year': 2025})
Process 2 received: ('MPI', {'type': 'Parallel Programming', 'year': 2025})
Process 3 received: ('MPI', {'type': 'Parallel Programming', 'year': 2025})
```

Question 8:

```
from mpi4py import MPI

comm = MPI.COMM_WORLD
rank = comm.Get_rank()

if rank == 0:
   comm.send("Confidential Message", dest=2)
elif rank == 2:
   data = comm.recv(source=0)
   print(f"Process {rank} received: {data}")
```

```
❯ mpirun -n 4 python 8.py
Process 2 received: Confidential Message
```

Question 9:

```
from mpi4py import MPI

comm = MPI.COMM_WORLD
rank = comm.Get_rank()

if rank == 0:
   data = [1, 2, 3, 4]
else:
   data = None

received = comm.scatter(data, root=0)
processed = received * received
result = comm.gather(processed, root=0)

if rank == 0:
   print("Squared Results:", result)
```

```
❯ mpirun -n 4 python 9.py
Squared Results: [1, 4, 9, 16]
```